

Question 7: [Programming, 10+10 Points]

Consider the Poisson equation

$$\begin{aligned}-\Delta u &= f \quad \text{in } \Omega, \\ u &= g \quad \text{on } \Gamma,\end{aligned}$$

where $\Gamma = \partial\Omega$ denotes the boundary of the domain.

We have derived the five-point stencil for solving the above PDE in the case $\Omega = [0, 1]^2$ with uniform grid spacing. In the attached Python script, you will find the code for this case.

The right-hand side f and the boundary condition g are chosen such that the exact solution is

$$u(x, y) = x^4 y^5 - 17 \sin(xy).$$

From theory, we expect a convergence rate of order 2, which can be verified numerically in the provided code.

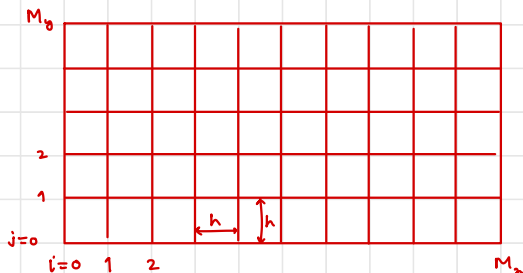
For bonus points, extend the implementation as follows:

- Modify the code so that the five-point stencil can be applied on a rectangular domain, e.g. $\Omega = [0, 2] \times [0, 1]$, using uniform grid spacing.
- Modify the code to allow for non-uniform grid spacing with mesh sizes h_x and h_y in the x - and y -directions, respectively.
- (Double Bonus Points)** Extend the code to handle the case of an L-shaped domain, i.e.,

$$\Omega = [0, 1]^2 \setminus \{[0.5, 1] \times [0.5, 1]\}.$$

For checking the solution in this case, you can use Example 2.17 from the lecture notes where f and g are chosen such that the exact solution is given by $u(x, y) = \sin(\pi x) \cos(\pi y)$.

② for rectangular uniform spaced grid,



such that, $i = 0, 1, 2, \dots, M_x, \quad j = 0, 1, 2, \dots, M_y$

Now, using 5-pt. stencil, we write Poisson eqⁿ for inner points at fix 'i'.

$$-I \bar{u}_{i-1} + D \bar{u}_i - I \bar{u}_{i+1} = h^2 \bar{f}_i + \bar{g}_i, \quad i=1,2,\dots, M_x-1$$

where, $I = I_{(M_y-1) \times (M_y-1)}$

$$D = \begin{bmatrix} 4 & -1 & 0 & \dots & 0 \\ -1 & 4 & -1 & 0 & \dots & 0 \\ 0 & -1 & 4 & -1 & 0 & \dots & 0 \\ \vdots & & & & & & \\ \vdots & & & & & & \\ 0 & \dots & \dots & \dots & -1 & 4 \end{bmatrix}_{(M_y-1) \times (M_y-1)}$$

$$\Delta \quad \bar{u} = \begin{bmatrix} u_{1,1} \\ u_{1,2} \\ \vdots \\ u_{1,M_y-1} \\ \hline \vdots \\ u_{i,1} \\ u_{i,2} \\ \vdots \\ u_{i,M_y-1} \\ \hline \vdots \\ u_{M_x-1,1} \\ u_{M_x-1,2} \\ \vdots \\ u_{M_x-1,M_y-1} \end{bmatrix}_{(M_x-1)(M_y-1) \times 1} = \underbrace{\begin{pmatrix} \bar{u}_i \end{pmatrix}}_{(M_y-1) \times 1}, \quad \bar{f}_i = \begin{bmatrix} f_{i,1} \\ f_{i,2} \\ \vdots \\ f_{i,M_y-1} \end{bmatrix}_{(M_y-1) \times 1}, \quad \bar{g}_i = \begin{bmatrix} g_{i,0} \\ 0 \\ 0 \\ \vdots \\ 0 \\ g_{i,(M_y-1)} \end{bmatrix}_{(M_y-1) \times 1}$$

Then stacking e_i^{ns} for $i = 1, 2, \dots, (M_x - 1)$, we get matrix e_i^{ns} .

$$A \bar{u} = \bar{f}$$

where,

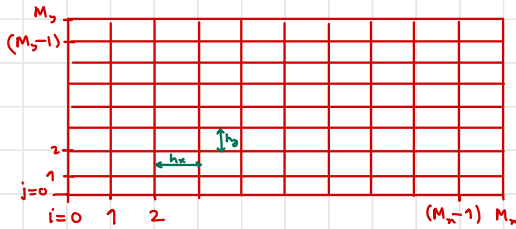
$$A = \frac{1}{h^2} \begin{bmatrix} D & -I & 0 & \dots & 0 \\ -I & D & -I & 0 & \dots & 0 \\ \vdots & & & & & \\ 0 & 0 & \dots & \dots & -I & D \end{bmatrix}_{(M_x-1)(M_y-1) \times (M_x-1)(M_y-1)}$$

$$\bar{f} = \begin{bmatrix} f_{1,1} + \theta_{1,0}/h^2 + \theta_{0,1}/h^2 \\ f_{1,2} + \theta_{0,2}/h^2 \\ \vdots \\ f_{1,M_y-1} + \theta_{1,M_y}/h^2 + \theta_{0,M_y-1}/h^2 \\ \vdots \\ f_{i,1} + \theta_{i,0}/h^2 \\ f_{i,2} \\ \vdots \\ f_{i,M_y-1} + \theta_{i,M_y}/h^2 \\ \vdots \\ f_{M_x-1,1} + \theta_{M_x-1,0}/h^2 + \theta_{M_x-1,1}/h^2 \\ f_{M_x-1,2} + \theta_{M_x-1,2}/h^2 \\ \vdots \\ f_{M_x-1,M_y-1} + \frac{\theta_{M_x-1,M_y}}{h^2} + \frac{\theta_{M_x,M_y-1}}{h^2} \end{bmatrix}_{(M_x-1)(M_y-1) \times 1}$$



See Python Notebook.

⑥ for rectangular non-uniform spaced grid



The Poisson eqⁿ in discrete form,

$$-\left(\frac{\partial^2 u}{\partial x^2} \Big|_{(i,j)} + \frac{\partial^2 u}{\partial y^2} \Big|_{(i,j)} \right) = f_{i,j}$$

Approximating, $\frac{\partial^2 u}{\partial x^2} \Big|_{(i,j)} \approx \frac{(u_{i+1,j} - 2u_{i,j} + u_{i-1,j}))}{h_x^2}$

And $\frac{\partial^2 u}{\partial y^2} \Big|_{(i,j)} \approx \frac{(u_{i,j+1} - 2u_{i,j} + u_{i,j-1}))}{h_y^2}$

$$\therefore 2u_{i,j} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{u_{i+1,j}}{h_x^2} - \frac{u_{i-1,j}}{h_x^2} - \frac{u_{i,j+1}}{h_y^2} - \frac{u_{i,j-1}}{h_y^2} = f_{i,j}$$

Now, using above, we can write Poisson eqⁿ for inner points at fix 'i',

for $j=1$, $2u_{i,1} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{u_{i+1,1}}{h_x^2} - \frac{u_{i-1,1}}{h_x^2} - \frac{u_{i,2}}{h_y^2} - \frac{u_{i,0}}{h_y^2} = f_{i,1}$

As, $u_{i,0} = \theta_{i,0}$ (boundary point)

$$\therefore 2u_{i,1} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{u_{i+1,1}}{h_x^2} - \frac{u_{i-1,1}}{h_x^2} - \frac{u_{i,2}}{h_y^2} = f_{i,1} + \frac{\theta_{i,0}}{h_y^2}$$

for $j = 2$,

$$2 U_{i,2} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{i+1,2}}{h_x^2} - \frac{U_{i-1,2}}{h_x^2} - \frac{U_{i,3}}{h_y^2} - \frac{U_{i,1}}{h_y^2} = f_{i,2} \quad \text{--- ②}$$

for $j = M_y - 1$,

$$2 U_{i,M_y-1} \left(\right) - \frac{U_{i+1,M_y-1}}{h_x^2} - \frac{U_{i-1,M_y-1}}{h_x^2} - \frac{U_{i,M_y}}{h_y^2} - \frac{U_{i,M_y-2}}{h_y^2} = f_{i,M_y-1}$$

As, $U_{i,M_y} = g_{i,M_y}$ (boundary point)

$$2 U_{i,M_y-1} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{i+1,M_y-1}}{h_x^2} - \frac{U_{i-1,M_y-1}}{h_x^2} - \frac{U_{i,M_y}}{h_y^2} = f_{i,M_y-1} + \frac{g_{i,M_y}}{h_y^2} \quad \text{--- ③}$$

Writing ①, ② & ③ in matrix form,

$$\textcircled{1} \equiv \begin{bmatrix} -\frac{1}{h_x^2} \\ \frac{1}{h_x^2} \end{bmatrix} [U_{i-1,1}] + \begin{bmatrix} 2 \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) & -\frac{1}{h_y^2} \end{bmatrix} \begin{bmatrix} U_{i,1} \\ U_{i,2} \end{bmatrix} + \begin{bmatrix} -\frac{1}{h_x^2} \end{bmatrix} [U_{i+1,1}] = f_{i,1} + \frac{g_{i,0}}{h_y^2}$$

$$\textcircled{2} \equiv \begin{bmatrix} -\frac{1}{h_x^2} \end{bmatrix} [U_{i-1,2}] + \begin{bmatrix} -\frac{1}{h_y^2} & 2 \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) \end{bmatrix} \begin{bmatrix} U_{i,1} \\ U_{i,2} \end{bmatrix} + \begin{bmatrix} -\frac{1}{h_x^2} \end{bmatrix} [U_{i+1,2}] = f_{i,2}$$

$$\textcircled{3} \equiv \begin{bmatrix} -\frac{1}{h_x^2} \end{bmatrix} [U_{i-1,M_y-1}] + \begin{bmatrix} -\frac{1}{h_y^2} & 2 \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) \end{bmatrix} \begin{bmatrix} U_{i,M_y-2} \\ U_{i,M_y-1} \end{bmatrix} + \begin{bmatrix} -\frac{1}{h_x^2} \end{bmatrix} [U_{i+1,M_y-1}] = f_{i,M_y-1} + \frac{g_{i,M_y}}{h_y^2}$$

Using the matrix form of ①, ② & ③, one can write $\varphi_j^{Ns} \forall j=1,2,\dots,M_3-1$

$$\left(-\frac{1}{h_m^2}\right) I \bar{u}_{i-1} + D' \bar{u}_i + \left(-\frac{1}{h_m^2}\right) I \bar{u}_{i+1} = \bar{f}_i + \bar{g}_i$$

where, $I = (M_3-1) \times (M_3-1)$ Identity matrix

$$D' = \begin{bmatrix} 2\left(\frac{1}{h_x^2} + \frac{1}{h_y^2}\right) & -\frac{1}{h_y^2} & 0 & \dots & 0 \\ -\frac{1}{h_x^2} & 2\left(\frac{1}{h_x^2} + \frac{1}{h_y^2}\right) & -\frac{1}{h_x^2} & 0 & \dots & 0 \\ 0 & -\frac{1}{h_x^2} & 2\left(\frac{1}{h_x^2} + \frac{1}{h_y^2}\right) & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -\frac{1}{h_y^2} & 2\left(\frac{1}{h_x^2} + \frac{1}{h_y^2}\right) \end{bmatrix}_{(M_3-1) \times (M_3-1)}$$

$$A \quad \bar{u} = \begin{bmatrix} u_{1,1} \\ u_{1,2} \\ \vdots \\ u_{1,M_3-1} \\ \hline \vdots \\ u_{i,1} \\ u_{i,2} \\ \vdots \\ u_{i,M_3-1} \\ \hline \vdots \\ u_{M_3-1,1} \\ u_{M_3-1,2} \\ \vdots \\ u_{M_3-1,M_3-1} \end{bmatrix}_{(M_3-1)(M_3-1) \times 1} = (\bar{u}_i)_{(M_3-1) \times 1}, \quad \bar{f}_i = \begin{bmatrix} f_{i,1} \\ f_{i,2} \\ \vdots \\ f_{i,M_3-1} \end{bmatrix}_{(M_3-1) \times 1}, \quad \bar{g}_i = \begin{bmatrix} g_{i,0}/h_y^2 \\ 0 \\ \vdots \\ 0 \\ g_{i,M_3}/h_y^2 \end{bmatrix}_{(M_3-1) \times 1}$$

Now, stacking e_j^{ns} for $j = 1, 2, \dots, (M_x - 1)$, we get matrix e_j^{ns} .

$$A' \bar{u} = \bar{f}'$$

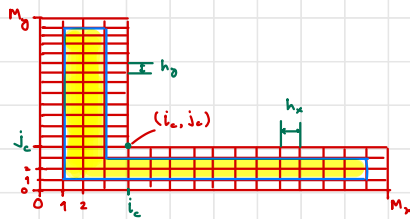
where,

$$A' = \begin{bmatrix} D' & -\frac{1}{h_x^2} I & 0 & \dots & 0 \\ \frac{1}{h_x^2} I & D' & -\frac{1}{h_x^2} I & 0 & \dots & 0 \\ \vdots & & & & & \\ 0 & 0 & \dots & \dots & -\frac{1}{h_x^2} I & D' \end{bmatrix}_{(M_x-1)(M_y-1) \times (M_x-1)(M_y-1)}$$

$$\bar{f}' = \begin{bmatrix} f_{1,1} + \theta_{1,0}/h_y^2 + \theta_{0,1}/h_x^2 \\ f_{1,2} + \theta_{0,2}/h_x^2 \\ \vdots \\ f_{1,M_y-1} + \theta_{1,M_y}/h_y^2 + \theta_{0,M_y-1}/h_x^2 \\ \vdots \\ f_{i,1} + \theta_{i,0}/h_y^2 \\ f_{i,2} \\ \vdots \\ f_{i,M_y-1} + \theta_{i,M_y}/h_y^2 \\ \vdots \\ f_{M_x-1,1} + \frac{\theta_{M_x-1,0}}{h_y^2} + \frac{\theta_{M_x,1}}{h_x^2} \\ f_{M_x-1,2} + \theta_{M_x,2}/h_x^2 \\ \vdots \\ f_{M_x-1,M_y-1} + \frac{\theta_{M_x-1,M_y}}{h_y^2} + \frac{\theta_{M_x,M_y-1}}{h_x^2} \end{bmatrix}_{(M_x-1)(M_y-1) \times 1}$$

→ See Python Notebook.

© General L-shape domain



In this grid, total no. of Unknown points are = $(M_x - 1)(j_c - 1) + (i_c - 1)(M_y - j_c)$

From previous sol., for any (i, j) pt. in non-uniform rectangular grid -

$$2U_{i,j} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{i+1,j}}{h_x^2} - \frac{U_{i-1,j}}{h_x^2} - \frac{U_{i,j+1}}{h_y^2} - \frac{U_{i,j-1}}{h_y^2} = f_{i,j}$$

for $j=1$ & $i=1$

Boundary pts.
 $U_{1,0} = \theta_{1,0}$
 $U_{0,1} = u_{0,1}$

$$2U_{1,1} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{2,1}}{h_x^2} - \frac{U_{1,2}}{h_y^2} = f_{1,1} + \frac{\theta_{1,0}}{h_y^2} + \frac{u_{0,1}}{h_x^2}$$

for $j=1$, $i=2, 3, \dots, M_x-2$

$$2U_{i,1} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{i+1,1}}{h_x^2} - \frac{U_{i-1,1}}{h_x^2} - \frac{U_{i,2}}{h_y^2} = f_{i,1} + \frac{\theta_{i,0}}{h_y^2}$$

for $j=1$, $i=M_x-1$

$$2U_{M_x-1,1} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{M_x-2,1}}{h_x^2} - \frac{U_{M_x-1,2}}{h_y^2} = f_{M_x-1,1} + \frac{\theta_{M_x-1,0}}{h_y^2} + \frac{u_{M_x-1,0}}{h_x^2}$$

Above 3 eqns in Matrix form -

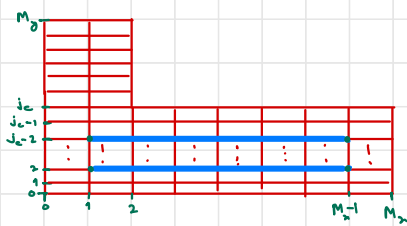
$$a_1 \bar{u}_1 = \bar{b}_1$$

where, $\bar{u}_1 \equiv (M_x-1) \times 1$, $a_1 \equiv (M_x-1) \times (M_x-1)$, $\bar{b}_1 \equiv (M_x-1) \times 1$

$$u_1 = \begin{bmatrix} u_{1,1} \\ u_{1,2} \\ \vdots \\ u_{1,M_n-1} \end{bmatrix}, \quad b_1 = \begin{bmatrix} f_{1,1} + \frac{g_{1,0}}{h_2^2} + \frac{g_{0,1}}{h_n^2} \\ f_{2,1} + g_{2,0}/h_2^2 \\ \vdots \\ f_{M_n-2,1} + g_{M_n-2,0}/h_2^2 \\ f_{M_n-1,1} + \frac{g_{M_n,1}}{h_2^2} + \frac{g_{M_n,0}}{h_2^2} \end{bmatrix} \quad \text{--- } (M_n-1) \times 1$$

And,

$$a_1 =$$



for $j=2, 3, \dots, j_c-2$, $i=1$

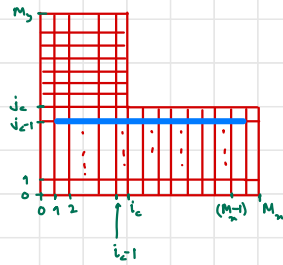
$$2U_{1,j} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{2,j}}{h_x^2} - \frac{U_{1,j+1}}{h_y^2} - \frac{U_{1,j-1}}{h_y^2} = f_{1,j} + \frac{g_{0,j}}{h_x^2}$$

for $j=2, 3, \dots, j_c-2$, $i=2, 3, \dots, M_x-2$

$$2U_{i,j} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{i+1,j}}{h_x^2} - \frac{U_{i-1,j}}{h_x^2} - \frac{U_{i,j+1}}{h_y^2} - \frac{U_{i,j-1}}{h_y^2} = f_{i,j}$$

for $j=2, 3, \dots, j_c-2$, $i=M_x-1$

$$2U_{M_x-1,j} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{M_x-2,j}}{h_x^2} - \frac{U_{M_x-1,j+1}}{h_y^2} - \frac{U_{M_x-1,j-1}}{h_y^2} = f_{M_x-1,j} + \frac{g_{M_x,j}}{h_x^2}$$



for $j = j_c - 1$, $i = 1$

$$2U_{1,j_c-1} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{2,j_c-1}}{h_x^2} - \frac{U_{1,j_c}}{h_y^2} - \frac{U_{1,j_c-2}}{h_y^2} = f_{1,j_c-1} + \frac{\theta_{0,j_c-1}}{h_x^2}$$

for $j = j_c - 1$, $i = 2, 3, \dots, i_c - 1$

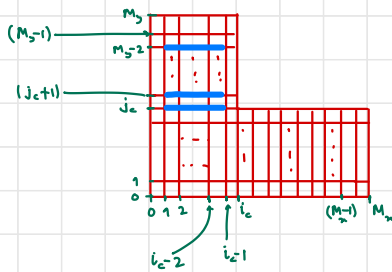
$$2U_{i,j_c-1} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{i+1,j_c-1}}{h_x^2} - \frac{U_{i-1,j_c-1}}{h_x^2} - \frac{U_{i,j_c}}{h_y^2} - \frac{U_{i,j_c-2}}{h_y^2} = f_{i,j_c-1}$$

for $j = j_c - 1$, $i = i_c, i_c + 1, \dots, M_x - 2$

$$2U_{i,j_c-1} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{i+1,j_c-1}}{h_x^2} - \frac{U_{i-1,j_c-1}}{h_x^2} - \frac{U_{i,j_c-2}}{h_y^2} = f_{i,j_c-1} + \frac{\theta_{i,j_c}}{h_y^2}$$

for $j = j_c - 1$, $i = M_x - 1$

$$2U_{M_x-1,j_c-1} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{M_x,j_c-1}}{h_x^2} - \frac{U_{M_x-2,j_c-1}}{h_x^2} = f_{M_x-1,j_c-1} + \frac{\theta_{M_x,j_c-1}}{h_x^2} + \frac{\theta_{M_x-1,j_c}}{h_y^2}$$



for $j = j_c, j_c+1, \dots, M_5-2$, $i = 1$

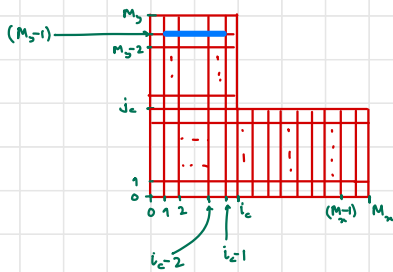
$$2U_{1,j} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{2,j}}{h_x^2} - \frac{U_{1,j+1}}{h_y^2} - \frac{U_{1,j-1}}{h_y^2} = f_{1,j} + \frac{g_{0,j}}{h_x^2}$$

for $j = j_c, j_c+1, \dots, M_5-2$, $i = 2, 3, \dots, i_c-2$

$$2U_{i,j} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{i+1,j}}{h_x^2} - \frac{U_{i-1,j}}{h_x^2} - \frac{U_{i,j+1}}{h_y^2} - \frac{U_{i,j-1}}{h_y^2} = f_{i,j}$$

for $j = j_c, j_c+1, \dots, M_5-2$, $i = i_c-1$

$$2U_{i_c-1,j} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{i_c-2,j}}{h_x^2} - \frac{U_{i_c-1,j+1}}{h_y^2} - \frac{U_{i_c-1,j-1}}{h_y^2} = f_{i_c-1,j} + \frac{g_{i_c-1,j}}{h_x^2}$$



for $j = M_y - 1, \quad i = 1$

$$2 U_{1, M_y-1} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{2, M_y-1}}{h_x^2} - \frac{U_{1, M_y-2}}{h_y^2} = f_{1, M_y-1} + \frac{\theta_{0, M_y-1}}{h_x^2} + \frac{\theta_{1, M_y}}{h_y^2}$$

for $j = M_y - 1, \quad i = 2, 3, \dots, i_c - 2$

$$2 U_{i, M_y-1} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{i+1, M_y-1}}{h_x^2} - \frac{U_{i-1, M_y-1}}{h_x^2} - \frac{U_{i, M_y-2}}{h_y^2} = f_{i, M_y-1} + \frac{\theta_{i, M_y}}{h_y^2}$$

for $j = M_y - 1, \quad i = i_c - 1$

$$2 U_{i_c-1, M_y-1} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) - \frac{U_{i_c-2, M_y-1}}{h_x^2} - \frac{U_{i_c, M_y-2}}{h_y^2} = f_{i_c-1, M_y-1} + \frac{\theta_{i_c-1, M_y-1}}{h_x^2} + \frac{\theta_{i_c-1, M_y}}{h_y^2}$$